

Spectroscopic analysis of organic contaminants in terephthalic acid: colour implications in poly(ethylene terephthalate) manufacture

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Colour measurements and doping experiments on polyethylene terephthalate (PET) show that the compounds in crude terephthalic acid which cause the yellow colour of derived PET polymer are carboxylic acid derivatives of anthraquinone and fluorenone. Fluorene derivatives are found to contribute to the PET polymer yellowness. On the other-hand, PET polymer luminance, was not affected by the small variations in impurity levels investigated in this work. 4-carboxybenzaldehyde (4-CBA) the major impurity in manufactured crude terephthalic acid (CTA), dominates the fluorescence spectrum. The fluorescence emission spectrum of 4-CBA is strongly influenced by the polarity and protic nature of the solvent. Associated dimers and aggregates of 4-CBA are primarily formed via the aldehyde and carboxylic acid groups causing marked red shifts in excitation and absorption spectra. Complexation also occurs between the 4-CBA and PTA groups. © 1998 Elsevier Science Limited. All rights reserved

1 INTRODUCTION

Poly(ethyleneterephthalate) (PET) is used extensively in the fibre and packaging industries. For most applications it is essential that the base polymer should exhibit only minor variations in physical/chemical properties, which results in uniform processability but additionally the optical properties luminance, yellowness and fluorescence must also be tightly controlled for aesthetic reasons and customer acceptance. Apart from fluorescence, two types of polymer colour variability can be identified. The first is characterised by an increase in yellowness while maintaining constant luminance, while the second exhibits a combined fall in both luminance and yellowness. The latter effect is usually associated with a reduction of antimony trioxide catalyst, used in the manufacture of the PET, the metallic antimony produced forming hexagonal plates clearly visible under microscopic examination. The effect is particularly noticeable if

a phosphorus (III) based stabiliser is used in the polymer recipe.¹ A similar effect is noted with dimethylterephthalate (DMT) based polymer where the transesterification catalyst is precipitated in an inactive form prior to polymerisation. In both cases the size of the particulate contamination is such that it alters the intensity and composition of the reflected light. Thus, terephthalic acid (TA) based polymers generally have increased luminance and yellowness when compared to similar polymers made from DMT.

The effect of polymer processing conditions on colour formation is also well-known,² consequently, when changes in polymer optical properties occur while operating at standard polymer processing conditions, a raw material quality (contamination) problem is indicated. Extensive studies have also been undertaken on the luminescence characteristics of PET with a view to ascertaining the nature of the oxidation products.³ Prolonged melt oxidation of PET results in extensive discolouration and the formation of highly fluorescent hydroxylated terephthalate units which exist in equilibrium with highly phosphorescent stilbenequinone units. Thus,

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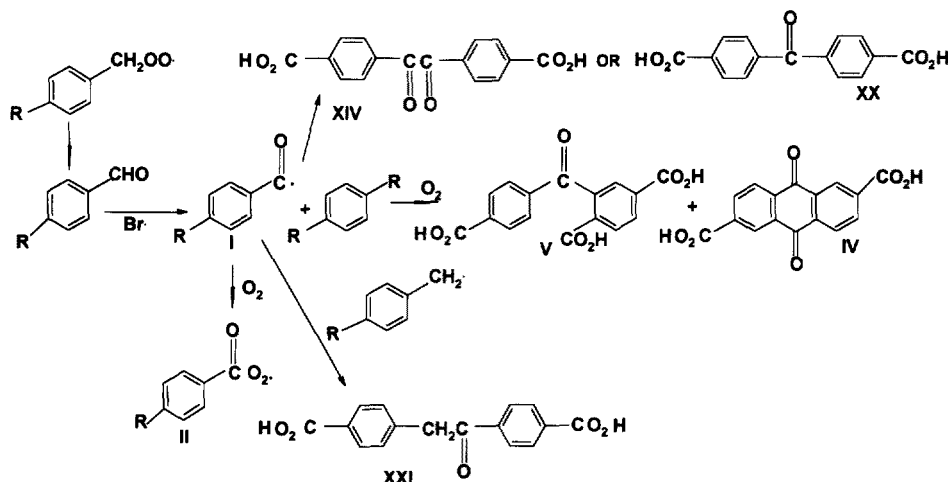
long-term reprocessing and recycling operations can give rise to additional colour problems.

Terephthalic acid is manufactured in a 2-stage process. In the first stage p-xylene is oxidised in acetic acid solution using a mixed catalyst of cobalt, manganese and bromide ions.⁴ The TA produced is 99.7% pure, the residual impurity being essentially the underconversion product, 4-carboxybenzaldehyde (4-CBA). However, the crude product produces an unacceptable yellow polymer with ethylene glycol. Further purification by hydrogenation over a palladium on carbon catalyst can give a 99.95% purity product.⁵ The chemical impurities in pure TA (PTA) are therefore potentially trace amounts of the original crude TA impurities together with their reduction products. The total concentration of the organic impurities may be monitored by color as measured by tristimulus colour meters, or by absorbance of a given concentration of an alkaline solution of PTA at a fixed wavelength usually 340 nm.

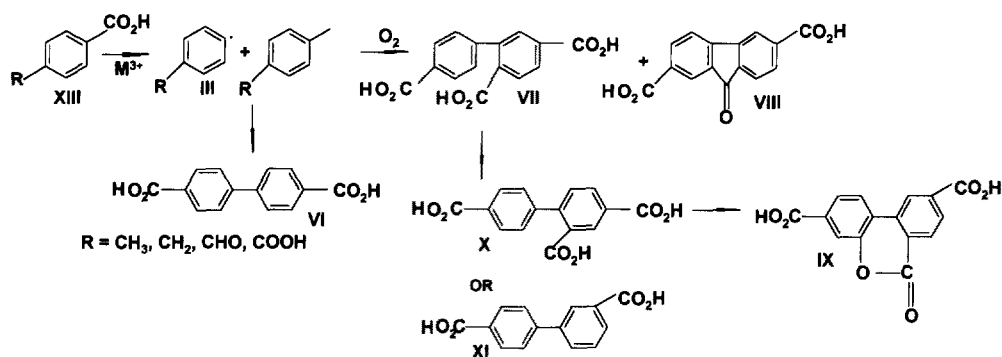
The colour forming impurities are believed to be due to compounds of the benzil, fluorenone, and

possibly anthraquinone type.^{5,6} Large varieties of groups of compounds have been identified by combined gas-chromatography/mass-spectrometry on the methyl ester derivatives.⁷ Substituted benzoic acid, phenol, biphenyl, benzophenone, fluorenone and anthraquinone were shown to be present but their relative concentrations were not discussed. Only limited information regarding TA impurities has been published.¹ A recent patent,⁸ however, has specified a number of the compounds identified by Roffia *et al.*⁷ as well as their reduction products as being present in TA mother liquors. We confirm that the major components of the deleterious trace impurities present in PTA are largely those specified by James⁸ which may be considered to arise as follows.

The oxidation stage is a free radical reaction and the reaction schemes shown (Schemes 1 and 2) conforms to that widely accepted for the auto-oxidation of hydrocarbons in solution. The efficiency of conversion to carboxylic acid is dependent on the aryl-oxy radical (I) having sufficient oxygen available to form the peroxy radical



Scheme 1. Formation of crude aromatic ketone structures in terephthalic acid manufacture.



Scheme 2. Formation of crude biphenyl and fluorenone structures in terephthalic acid manufacture.

(II). If not, the aryl-oxy radical may either decarbonylate to an aryl radical (III) or it may attack a substrate molecule to yield an orthosubstituted terephthalic acid derivative which may subsequently either cyclise and oxidise to produce ultimately anthraquinone-2, 6-dicarboxylic acid (IV) or oxidise to benzophenone-2,4',5-tricarboxylic acid (V). The aryl radical (III) yields benzoic acid derivatives, biphenyl derivatives (VI) by ipso attack,⁹ or ortho-substituted terephthalic acid derivatives (VII) which may cyclise and oxidise to fluorenone-2,6-dicarboxylic acid (VIII). We find the concentration of impurities derived from aryl radicals is greater than that of the impurities derived from aryl-oxy radicals presumably due to the increased concentration of these radicals generated by oxidative decarboxylation (XIII-III). The 3,7-dicarboxybenz(c)coumarin (IX) identified as present in TA mother liquor⁸ may arise from the radical (X) which can either cyclise¹⁰ or decarboxylate to give the biphenyl-3,4'-dicarboxylic acid (XI) also identified as present in the same stream. Minor amounts of compounds such as the benzil (XII) may arise via radical recombination reactions. In addition, trace impurities with molecular weights up to 490 have been identified in crude terephthalic acid process streams by Tandem mass spectroscopy on methyl ester derivatives. Mass spectral cracking patterns suggest that these high molecular weight materials are in all cases polycarboxylic acids containing three phenyl substituents some of which have cyclised to phenyl and benzoyl substituted fluorenone and anthraquinone carboxylic acids (Scheme 3). The concentration of these latter impurities is at least an order of magnitude lower than the dimeric impurities outlined

in (Scheme 1) whose individual concentrations are generally very much less than 100 ppm. The dimeric impurities and their reduction products have been synthesised in this work and their optical, spectroscopic and colour properties investigated on doped TA and polymer materials. The spectroscopic properties of the major chromophore responsible for their formation, 4-CBA, are also investigated in some detail via fluorescence and derivative UV/visible absorption techniques as potential analytical probes.

2 EXPERIMENTAL

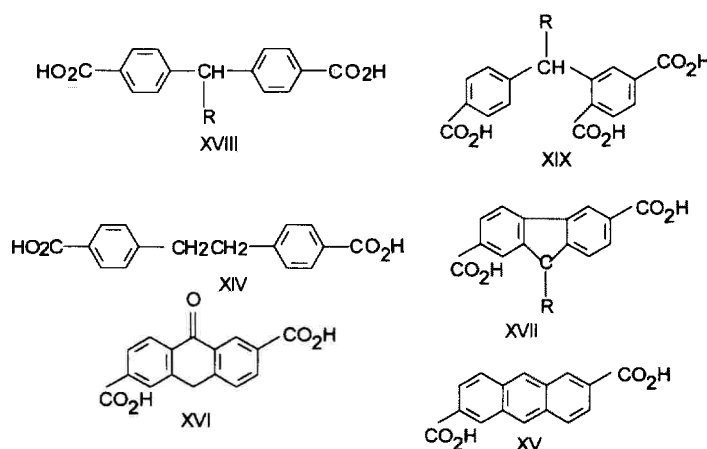
2.1 Materials

All compounds and solvents (spectroscopic grade) used in this study were obtained from the Aldrich Chemical Company, UK and used as supplied. The sulphuric acid used in this study was 98% v/v concentrated.

2.2 Yellowness Y

This was recorded using a Digital Colour Meter designed and manufactured by ICI Engineering Ltd. The yellowness is recorded as a function of reflected green and red light a over narrow range of chromaticity, the units of yellowness being adjudged equal to unit shade difference noticeable to the eye. The yellowness Y correlates to CIE b^* (measured by Colorgard 2000/05) over the range covered as:

$$b^* = -0.78 + 0.19 Y \text{ for PTA}$$



Scheme 3. Crude terephthalic acid impurity reduction products.

2.3 Polymer preparation

Additives were added at the direct esterification stage as the free carboxylic acids. After removal of water (99%), the required amount of a standard batch of filtered antimony glycollate solution in ethylene glycol was added as polyesterification catalyst, and phosphoric acid as stabiliser, then the pressure was reduced under controlled conditions and polymerisation carried out to a standard viscosity. The resultant polymers were analysed for catalyst residues, end group concentrations, intrinsic viscosity, and diethylene glycol content to ensure any variations in optical properties noted were due to additives and not abnormal processing. Detailed results are noted in Table 1.

2.4 UV at 340 nm

The determination of UV absorbance of terephthalic acid is based upon a 10% w/v solution in 2 M.KOH. Solutions of the individual compounds in 2 M.KOH were scanned in a 1 cm cell from 450–310 nm on a Hitachi W-spectrophotometer. The absorbance at 340 nm of each compound at a particular concentration was measured and the concentration in PTA required to increase absorbance by 0.01 absorbance units calculated.

2.5 Fluorescence analysis

Fluorescence excitation and emission spectra were recorded using a Perkin–Elmer LS50B lumines-

cence research spectrometer. All spectra were recorded in aerobic solutions at room temperature conditions.

2.6 HPLC analysis

Three per cent solutions produced by neutralising PTA in 2 M.NaOH were analysed using a Spherisorb ODS-2 column eluting with an aqueous mobile phase which comprised tetrabutylammonium hydroxide (6.6 g 40% solution neutralised to pH 7 with phosphoric acid) and n-pentanol (20 ml) per litre. Responses at 240 and 275 nm were recorded with a Pye Unicam P.U.4020 W detector. A Shimadzu RF 530 fluorescence detector was used for fluorene-2,6-dicarboxylic acid (Ex.285 nm/Em.335 nm), and anthracene-2,6-dicarboxylic acid (Ex.273 nm/Em.440 nm) determinations.

The anthraquinone-2,6-dicarboxylic acid content of crude or pure terephthalic acid was determined as follows: TA (1.5 g) with and without added known amounts of anthraquinone-2,6-dicarboxylic acid, in ammonia solution (75 cm³; 880 water) was refluxed with zinc powder (7.5 g) for 4 h then poured into water (150 cm³). The excess zinc was removed by filtration and the filtrate acidified. The precipitated acid was filtered off then dissolved in caustic solution prior to analysis as above. The anthraquinone-2,6-dicarboxylic acid content of PET polymer was determined by prior hydrolyses in alkaline solution followed by acidification then following the above method.

Table 1. TA organic impurities: concentration range, colour and effect on pet optical properties

Compound	Concentration range in TA (ppm)	Colour	ppm in TA equivalent to 0.01 absorbance at 340nm	Amount added PET (ppm)	Derived polymer colour change	
					Luminance	Yellowness
p-Toluic acid	80–140	White		400	None	None
Benzoic acid	< 10	White		400	None	None
Isophthalic acid	< 10	White		400	None	None
4-Carboxybenzaldehyde	<20	White	100	400	None	None
Fluorene-2,6-dicarboxylic acid (XVII)	5–40	White (F)	27	10–50	None	1Y/120ppm
Anthracene-2,6-dicarboxylic acid (XV)	0.06–0.8	Yellow (F)	8	0.1–1.0	None	None
Trimellitic acid	20–40	White	10000	50–150	None	None
Biphenyl-2,4',5-tricarboxylic acid (VII)	20–50	White	22.9	10–50	None	None
4,4'-Diphenic acid (VI)	60–120	White	1250	400	None	None
Anthraquinone-2,6-dicarboxylic acid (IV)	<0.5	Green-yellow	6.3	1–5	None	1Y/1ppm
Fluorenone-2,6-dicarboxylic acid (VIII)	n/d	Intense yellow	15	10–50	None	1Y/10ppm
Bibenzyl-4,4'-dicarboxylic acid (XVI)	n/d	White	1400	400	None	None
Benzil-4,4'-dicarboxylic acid (XII)	n/d	Pale yellow	200	20	None	None
Benzophenone-4,4'-dicarboxylic acid (XX)	n/d	White	44	400	None	None
Benzophenone-2,4',5-tricarboxylic acid (V)	n/d	White	320	25	None	None
Biphenylmethane-4,4'-dicarboxylic acid (XVIII)	n/d	White	2900	20	None	None
Biphenylmethane-2,4',5-tricarboxylic acid (XIX)	n/d	White	1180	20	None	None
Anthrone-2,6-dicarboxylic acid (XVI)	n/d	Tan	5.6	1	None	None

2.7 Synthesis undertaken

2.7.1 Biphenyl-2,4',5-tricarboxylic acid (VII)

2,4',5-Trimethylbiphenyl (21.5 g) prepared by the method of Nealy and Adams was oxidised by potassium permanganate (120 g) added portion wise over 3 h in refluxing aqueous pyridine solution (150:450 cm³). After removal of excess oxidant with methanol followed by filtration then removal of pyridine the solution was acidified to precipitate the crude acid (22.5 g). Extraction with warm acetone afforded essentially pure 2,4',5-biphenyl-tricarboxylic acid. Trimethyl ester had an m.p. 107–108°C (ex cyclohexane). Lit. 105–108°C.

2.7.2 Fluorenone-2,6-dicarboxylic acid (VIII)

Finely ground biphenyl-2,4',5-tricarboxylic acid (5 g) was added with stirring to a solution of phosphorus pentoxide (50 g) in orthophosphoric acid S.G. (50 cm³) and the temperature raised to 150–160°C. After 1 h the resulting yellow suspension was cooled, poured into water and the crude fluorenone-2,6-dicarboxylic acid (2.9 g) removed by filtration then washed with water and acetone. Dimethyl ester had an m.p. 195–196°C (ex methanol).

2.7.3 Dimethyl-6-oxo-dibenzo(b,d)pyran-3,9-dicarboxylate (IX)

Hydrogen peroxide (5 cm³, 30%) was added to a solution of dimethylfluorenone-2,6-dicarboxylate (5 g) in trifluoroacetic acid and the mixture refluxed for 1 h after which a further amount of hydrogen peroxide (10 cm³, 30%) was added and refluxing continued until no more oxidant remained. On pouring into water a cream solid (4.3 g) was obtained which was recrystallised from ethyl acetate. NMR analysis indicated both potential isomers present the desired isomer accounting for 80% of the total.

2.7.4 Fluorene-2,6-dicarboxylic acid (XVII)

Fluorenone-2,6-dicarboxylic acid (2.7 g), potassium hydroxide (1.7 g), hydrazine hydrate (1.75 ml), and diethylene glycol (50 cm³), were heated together under gentle reflux for 4 h. On pouring into water and acidifying a gelatinous precipitate of fluorene-2,6-dicarboxylic acid (2.5 g) was obtained. Dimethyl ester had an m.p. 125°C (ex chloroform/methanol).

2.7.5 Anthracene-2,6-dicarboxylic acid (XV)

Anthraquinone-2,6-dicarboxylic acid (2.0 g) was stirred and refluxed with ammonium hydroxide

(100 cm³, 0.880), and powdered zinc (10 g) for 4 h. The resulting suspension was cooled, filtered and the resulting filtrate acidified to precipitate bright yellow anthracene-2,6-dicarboxylic acid (1.6 g). Dimethyl ester had an m.p. 272–273°C (ex chloroform).

2.7.6 Anthrone-2,6-dicarboxylic acid Dimethyl ester (XVI)

Sodium borohydride (2.0 g) was added portion wise over 1 h to a stirred suspension of dimethyl anthraquinone-2,6-dicarboxylate (2 g) such that the temperature remained below 0°C. The suspension was allowed to warm to room temperature overnight then diluted with water and filtered. The solid was extracted with boiling methanol (50 cm³) and unreacted anthraquinone filtered off. Removal of solvent afforded colourless crystals of dimethylantraquinol-2,6-dicarboxylate (1.5 g) m.p. 213–215°C (ex methanol). Dimethyl anthraquinol-2,6-dicarboxylate (1.0 g) stirred with 5 M. hydrochloric acid at room temperature overnight yielded the orange red solid dimethyl anthrone-2,6-dicarboxylate (0.9 g) m.p. 219–222°C (ex methanol).

2.7.7 4,4'-DimethoxycarbonylBenzil Dimethyl ester (XIV)

1,2-Di-(4-methoxycarbonylphenyl)ethane-1,2-diol was prepared by adding titanium tetrachloride (17.2 g) to a stirred solution of 4-methoxycarbonylbenzaldehyde (9.8 g) in dry tetrahydrofuran (40 cm³) in an argon atmosphere. The thick brown suspension was cooled to 0°C then powdered zinc (12 g) was added portionwise maintaining the temperature at < 5°C. The reaction mixture was stirred a further 2 h at 0°C. After the addition was completed the reaction mixture was neutralised with 10% potassium carbonate solution and extracted with ether. After drying and removal of solvent 10 g of white solid was obtained. FT nmr indicated the solid was largely the desired compound contaminated with a little 4,4'-dimethoxycarbonylstilbene. A portion of the diol (2.72 g) was oxidised to the bright yellow benzil which crystallised out of the oxidising mixture of dimethylsulphoxide (29 cm³) and acetic anhydride (16 cm³) overnight.

3 RESULTS AND DISCUSSION

3.1 Polymer colour

The effects of known and potential trace impurities in PTA defined above, on the optical properties of

a derived clear polymer prepared on a semi-technical scale are recorded in Table 1. It is assumed that the response of the major component of each type of compound is typical of the group to which it belongs and is only indicative of trends. Considerable experimentation on each additive would be necessary before quantitative statements could be made and these would require further refinements to define the effect of scale up to manufacturing conditions. Polymers produced on a semi-technical scale are exposed to high processing temperatures for much shorter periods than polymers produced commercially. For this reason additives were added at levels well above those normally found in PTA in order to exacerbate any effects on derived polymer colour.

The results in Table 1 clearly show some classes of compounds increase the yellowness of clear polymer although no concurrent changes in luminance are observed. Anthraquinone-2,6-dicarboxylic acid (IV) is a particularly virulent colour former, one ppm (based on PTA) producing on average an increase of one yellowness unit in the derived polymer. PTA contains <0.5 ppm of this compound but crude TA may contain up to 10 ppm. The increased polymer yellowness is possibly due either to the formation of the semi-quinone radical intermediate, which can be either fixed into the polymer backbone or stabilised by the rigid matrix (from reaction with oxygen)¹¹ or to the formation of a dihydroxy dianion salt.¹² However, the anthrone derivative (XVI), which also forms coloured salts failed to cause any noticeable change in derived polymer optical properties.

Fluorene-2, 6-dicarboxylic acid (XVII, R=H) increased the colour of derived polymer on average by one yellowness unit for every 10 ppm (based on PTA) added. Analysis of the recovered carboxylic acids after hydrolysis of the polymer revealed only 20% of the fluorene is accounted for, suggesting that a derivative is responsible for the increased yellowness. Oxidation back to the fluorenone precursor (VIII) did not occur. Possibly acetaldehyde, which is continuously generated during the polymerisation process, is condensing with the acidic methylene group to form a coloured fluorilidene derivative. Fluorenone-2,6-dicarboxylic acid (VIII), which is intense yellow in solution, increased the polymer colour as expected, but at the level found in PTA this compound does not contribute significantly to PET colour variations. In situ generation from biphenyl-2,4',5-tricarboxylic acid (VII) via acid catalysed cyclisation

did not occur. Benzil-4,4'-dicarboxylic acid (XII) had no detectable effect on derived polymer optical properties at levels up to 20 ppm (based on PTA), and neither did the addition of anthracene-2,6-dicarboxylic acid (XV) at levels up to 1.0 ppm (based on PTA). However, a solution of anthracene-2,6-dicarboxylic acid (1.0 ppm) dissolved in ethylene glycol exhibits clear blue fluorescence when irradiated with UV light. A similar strength solution in amorphous PET polymer should show a similar effect, however, no difference in relative fluorescence was observed when doped polymers were compared with controls.

The contribution of the various impurities to the absorbance at 340 nm is shown in Table 1 as the calculated concentration in PTA required to produce an increase of 0.01 absorbance units. Clearly anthracene, anthraquinone, fluorenone and fluorene derivatives are major contributors to increased absorbance of PTA solutions at 340 nm. Phenyl substitution onto these chromophores will increase conjugation displacing the absorption bands to longer wavelengths together with an increase in intensity.¹³ Since the UV absorbance at 340 nm is a solution test it reflects the total concentration of impurities present. Tristimulus measurements however, correspond to the total concentration of impurities present on the solid surface of the powder, which may or may not be representative of bulk concentration. In addition, those compounds responding to tristimulus irradiation to measure yellowness are not the most active on irradiation at 340 nm. Consequently, the two tests (solution and powder) do not therefore, necessarily correspond but, rather compliment each other. Both test, however, are extremely sensitive to small changes in these types of impurities, which the results above show can be responsible for colour variations in PET. The PTA manufacturer therefore, controls the product colour within a narrow predetermined range.⁶

3.2 Effect of PTA colour on PET colour

The results recorded in Table 1 show that no significant variation in optical properties ('Y' and 'L') is detected from the use of in house PTA with colour varying between +0.9 to -3Y units on derived polymer prepared on the semi-technical scale. Whereas, on the manufacturing scale, utilising batch polymerisation, no significant change in derived polymer colour was detected from the use of PTA varying in colour over the range +0.5 to

–2.5 Y units. The normal range of colour of commercial PTA is +1.0 to –1.5Y units. These particular PTA samples varied in percentage transmission at 340 nm over the range 90.5–94.3% but as discussed previously no correlation should be attempted between the two optical measurements.

The range of impurities present in the various sources of PTA does vary however, as shown in Table 1. This is because the final PTA product quality will depend upon a number of factors such as crude terephthalic acid quality, the severity of hydrogenation conditions to which the crude TA is subjected, mother liquor impurity concentration, rate of crystallisation, product separation temperatures and the efficiency of product washing. Consequently, each individual manufacturers process will have these individual stages optimised to different circumstances.

3.3 Spectral fluorescence characteristics of terephthalic acid impurities

As indicated above, the relationship between coloured impurities and luminescence is not necessarily synonymous. PET is known to be highly luminescent (fluorescent and phosphorescent)³ with the nature of the chromophores continuing to be a subject of some controversy with the manufacturing and processing history of the polymer playing a crucial role. Thus, the purity of the original TA used in the manufacture of the PET is important and any carry over of the various transformation products discussed above (Table 1) could be important in this regard. One of the major components in this regard is 4-carboxybenzaldehyde (4-CBA) which can undergo further reactions during the polymerisation process. Indeed, the fluorescence behaviour of 4-CBA in this work, although spectrally different, is not unlike that of PET itself.

The fluorescence of 4-CBA was found to be very dependent upon the solvent. Thus, in dimethylformamide (DMF), 4-CBA was weakly soluble but only very weakly fluorescent. Comparisons with a crude terephthalic acid (CTA) sample are shown in Fig. 1. The CTA is seen to exhibit a very broad structureless emission spectrum centred at 380 nm. Doping the CTA with 4-CBA gives an increased structured emission spectrum centred at 383 nm. However, strangely, this emission spectrum of 4-CBA in DMF decreases with increasing concentration. There is evidently some type of interaction between

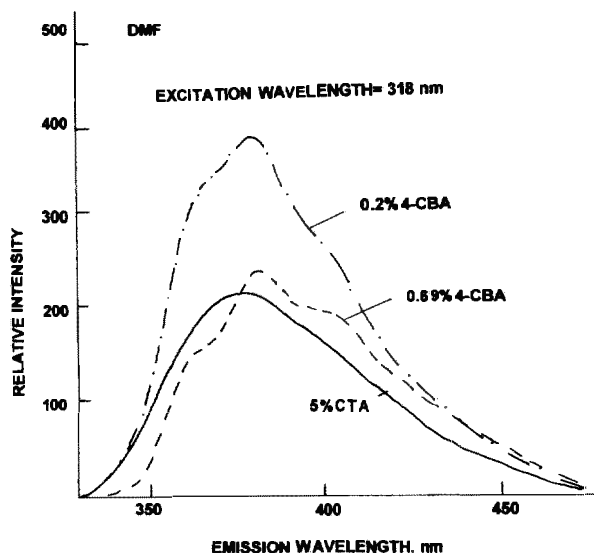


Fig. 1. Fluorescence emission spectra of (—) 5% w/v CTA with (-----) 0.69% w/v 4-CBA and (- · - · -) 0.2% w/v 4-CBA in DMF.

the TA in CTA and the 4-CBA since the latter exhibit very weak emission alone in DMF (< 10 on the same intensity scale at the same concentration).

Increased fluorescence was observed, on the other-hand, in more polar solvents such as alcohols. The spectral characteristics of 4-CBA in methanol are shown in Fig. 2. In this case two distinct fluorescence emission spectra are observed centred at 415 and 480 nm depending upon the excitation wavelength. Both spectra are observed over a wide concentration range. In this example, at 0.45% w/v two excitation bands are observed; a main band centred at 340 nm and a shoulder at

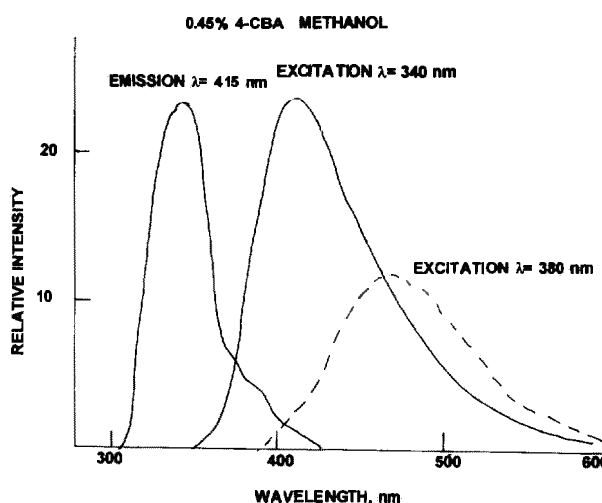


Fig. 2. Fluorescence excitation and emission spectra (—) (EX λ = 340 nm, EM λ = 415 nm) and emission spectrum (-----) (EX λ = 380 nm) of 0.45% w/v 4-CBA in methanol.

380 nm. This indicates the presence of at least two distinct emitting chromophoric species. However, in methanol the fluorescence was still relatively weak requiring high concentrations of 4-CBA for analysis. In order to enhance the sensitivity of the method only strongly protic solvents were viable such as the concentrated acids, acetic and sulphuric. In concentrated sulphuric acid intense fluorescence is observed centred at 460 nm and is independent of the excitation wavelength (Fig. 3). At the concentration of 0.5% w/v used here an intense excitation band is observed at 384 nm with a weak shoulder at 323 nm.

The most interesting feature of the 4-CBA fluorescence characteristics is the concentration dependence of the excitation wavelength. This feature is shown in Fig. 4 where there is a marked blue shift in the excitation spectra with decreasing concentration of 4-CBA. At the lowest concentration of 0.002% w/v three excitation bands are observed centred at 250, 300 and 340 nm. These spectral shifts are also reflected in the corresponding absorption spectra of 4-CBA.

The normal absorption spectra of 4-CBA in methanol and sulphuric acid are compared in Fig. 5. Whilst the absorption profiles are similar the spectral maxima are markedly red-shifted from methanol to sulphuric acid. The second order derivative UV spectra in Fig. 6 show the effect of changing concentration of 4-CBA. The long wavelength absorption maxima at high concentration gradually blue shifts and disappears at lower

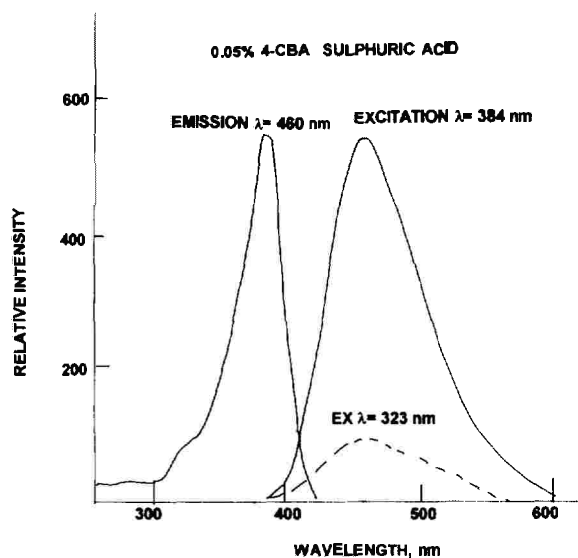


Fig. 3. Fluorescence excitation and emission spectra (—) (EX $\lambda = 384$ nm, EM $\lambda = 460$ nm) and emission spectrum (-----) (EX $\lambda = 323$ nm) of 0.05% w/v 4-CBA in Conc. H_2SO_4 .

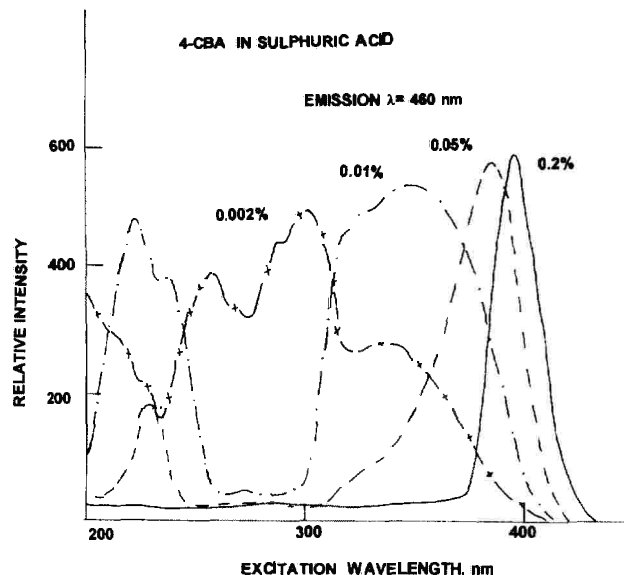


Fig. 4. Fluorescence excitation spectra (EM $\lambda = 460$ nm) at (—) 0.2%, (-----) 0.05%, (— · — · —) 0.01% and (x-x-x-x) 0.002% w/v 4-CBA in Conc. H_2SO_4 .

concentrations with new shorter wavelength absorption bands appearing.

Fluorescence analysis of pure TA shows the presence of an emission band at 383 nm which does not shift with excitation wavelength (Fig. 7). Crude TA (CTA) also gives rise to the same emission band but at longer excitation wavelengths (390 nm) gives rise to a second emission band at 460 nm. This second emission component is evidently due to 4-CBA as shown by the observation of enhanced emission at the same wavelength on doping with 4-CBA. However, the shorter wavelength emission at

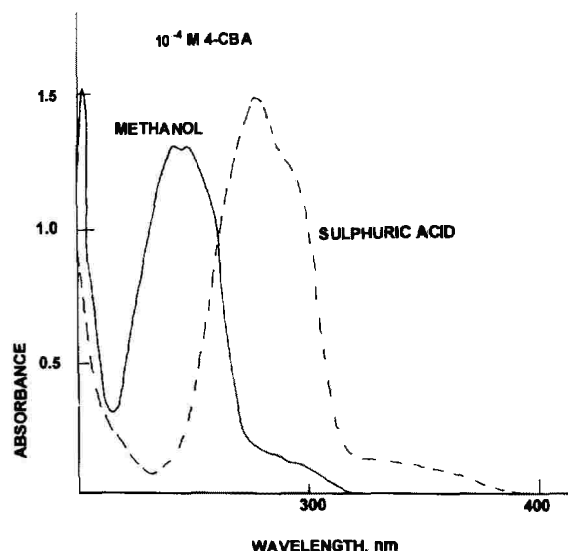


Fig. 5. Absorption spectra of 4-CBA (10^{-4} M) in (—) methanol and (-----) sulphuric acid.

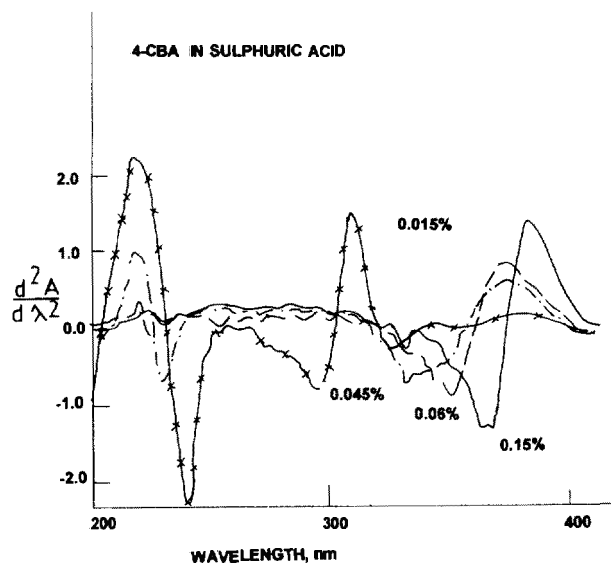


Fig. 6. Second order derivative UV absorption spectra of 4-CBA in H_2SO_4 at (—) 0.15%, (---) 0.06%, (.....) 0.045% and 0.015% w/v.

383 nm associated with the PTA is reduced with increasing additions of 4-CBA. Thus, the 4-CBA and the TA must be interacting in some way via complex formation. Calibration plots for the observed emission bands at 383 and 460 nm are shown in Fig. 8. Above 0.2% w/v 4-CBA, the emission at 460 nm achieves a plateau.

The observed spectral shifts for 4-CBA in more polar methanol and sulphuric acid media may be due to a number of possible reactions, all of which are reversible. The first, is due to the formation of associated molecular dimers or aggregates of the 4-CBA aldehyde and carboxylic acid groups as

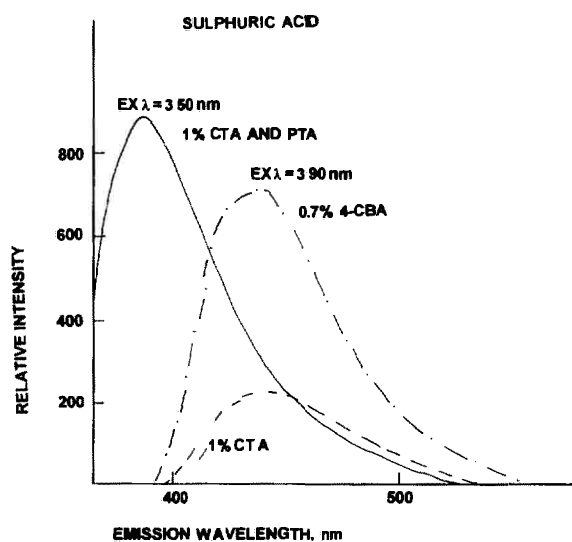


Fig. 7. Fluorescence emission spectra of (—) 1% CTA and PTA ($EX \lambda = 350$ nm), (---) 1% CTA ($EX \lambda = 390$ nm) and (.....) 0.7% w/v 4-CBA in Conc. H_2SO_4 .

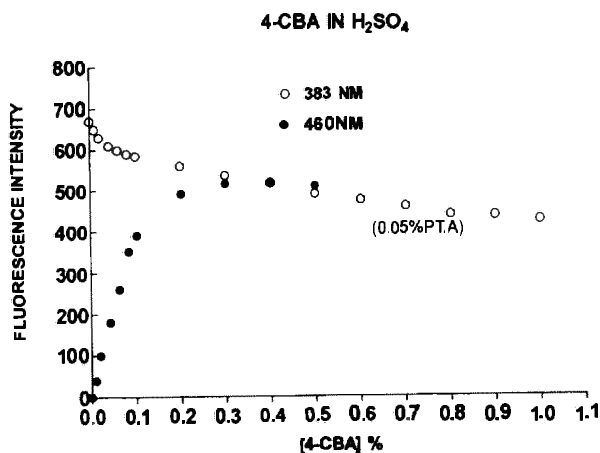
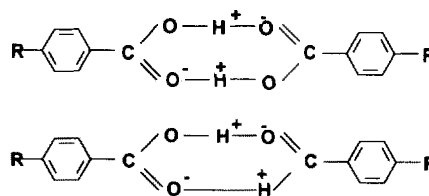


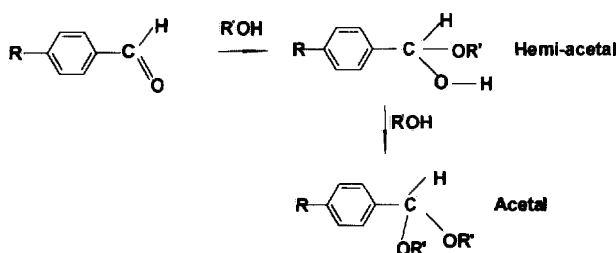
Fig. 8. Plots of fluorescence emission intensity versus concentration (%w/v) of 4-CBA at 460 and 383 nm in the presence of 0.05% w/v PTA.

shown in Scheme 4. The presence of such species would certainly account for the observed blue shifts in excitation wavelength maxima and derivative UV spectra with decreasing concentration. The second type of reaction involves the formation of a hemi-acetal or acetal species as shown in Scheme 5. These species would account for the red-shifted emission bands in methanol solution where the alcohol group adds to the aldehyde. In the third reaction, the presence of strong acid can result in protonation and hydration of the aldehyde group as shown in Scheme 6. The strong acid would maintain the equilibrium to the right to give the diol species.

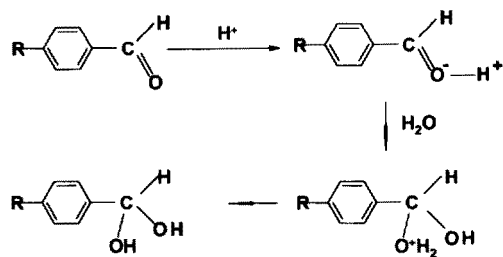
The fluorescence spectra of analogues of 4-CBA are shown in Figs 9 and 10 in sulphuric acid media. Terephthalaldehyde was found to be fluorescent in



Scheme 4. Formation of dimers/aggregates.



Scheme 5. Hemi-acetyl/acetyl formation.



Scheme 6. Protonation/hydration.

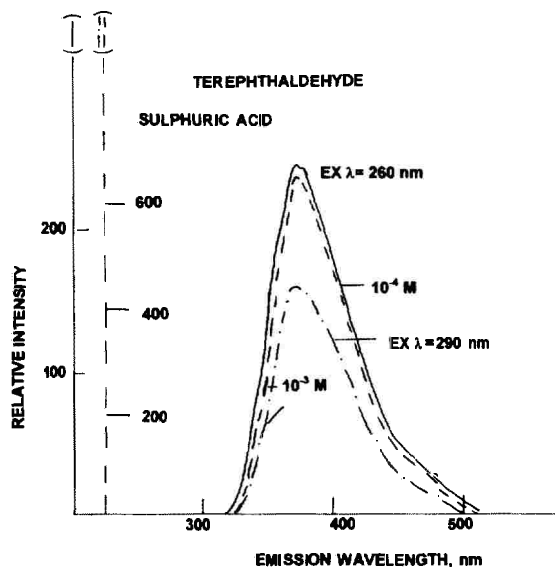


Fig. 9. Fluorescence emission spectra of (—) (10^{-4} M) and (---) (10^{-3} M) terephthalaldehyde (EX $\lambda = 260$ nm) and (---) (EX $\lambda = 290$ nm) in Conc. H_2SO_4 .

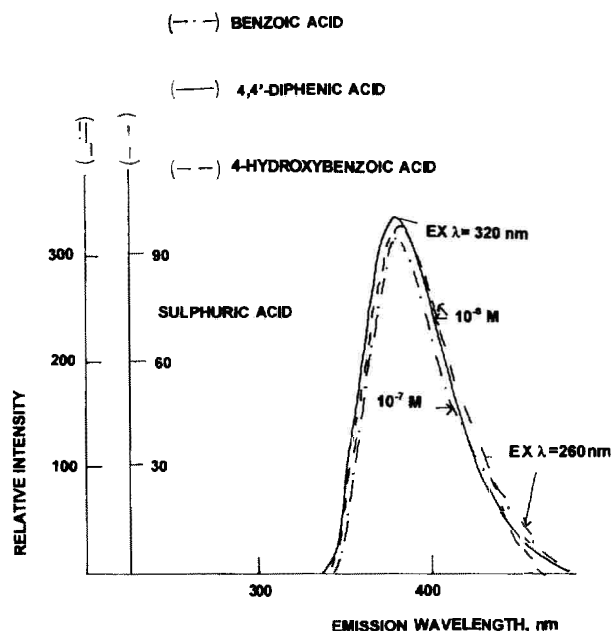


Fig. 10. Fluorescence emission spectra of (—) (10^{-6} M) 4,4'-diphenic acid and (---) 4-hydroxybenzoic acid (10^{-6} M) (EX $\lambda = 320$ nm) and (---) benzoic acid (EX $\lambda = 260$ nm) in Conc. H_2SO_4 .

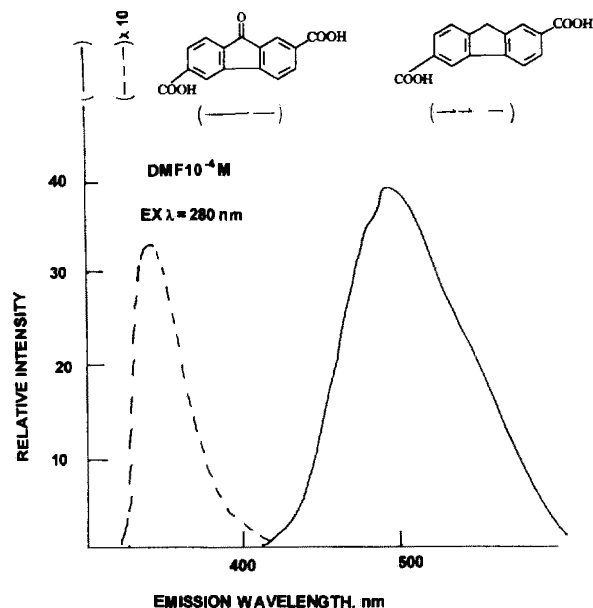


Fig. 11. Fluorescence emission spectra of (—) 2,6-dicarboxyfluorenone (10^{-4} M) and (---) 2,6-dicarboxyfluorene (10^{-4} M) (EX $\lambda = 280$ nm) in DMF.

sulphuric acid and has an emission maximum at 383 nm. This emission spectrum did not shift with changing concentration or excitation wavelength as shown (Fig. 9). Benzoic acid, 4-hydroxybenzoic acid and 4,4'-diphenic acid all gave similar emission spectra at 384 nm. No corresponding spectral shifts were observed with changing concentration or excitation wavelength. Thus, under these circumstances it would appear that the spectral shifts for the 4-CBA are primarily associated with the formation of associated dimers or aggregates. In the presence of TA these aggregates are broken up by competition with the carboxylic acid groups of the TA.

Finally, the fluorescence emission spectra of the coloured 2,6-dicarboxyfluorene and fluorenone derivatives are shown in Fig. 11. The former exhibits an emission at 330 nm while the latter emits at 500 nm. Neither of these emissions were observed in the CTA supplied. Anthraquinone-2,6-dicarboxylic acid was also examined but does not fluoresce.

4 CONCLUSIONS

The doping data shows that the compounds in crude terephthalic acid which cause the yellow colour of derived PET polymer are carboxylic acid derivatives of anthraquinone and fluorenone. These compounds, are essentially eliminated by the

purification process. Anthraquinone derivatives and their reduction products are largely removed in the aqueous mother liquor whereas fluorenones are converted to fluorenes, a proportion of which co-crystallise with the PTA. Fluorene derivatives also contribute to the PET polymer yellowness. On the other-hand, PET polymer luminance, was not affected by the small variation in impurity levels investigated in this work.

The combined total contribution of the impurities present in PTA may be evaluated by the UV absorbance test at 340 nm or a Tristimulus colour meter. The concentration of the possible individual components of the mixture will depend on how the manufacturer has optimised the process. Consequently, PTA obtained from different sources may show slight differences in PET polymer colour even though the base materials have similar colour characteristics. Since UV absorbance at 340 nm is a solution test and Tristimulus is measured on powders it is possible to obtain PTA with low yellowness but relatively poor transmission characteristics and vice versa.

Fluorescence analysis of the CTA showed the presence of 4-CBA as a primary impurity component. The emission spectra of 4-CBA is strongly influenced by the polarity and protic nature of the solvent. Associated dimers and aggregates of 4-CBA are formed via the aldehyde and carboxylic acid groups causing marked shifts in excitation and absorption spectra.

Complexation also occurs between the 4-CBA and PTA groups. Finally, it is interesting to note the fact that previously reported fluorescence

characteristics of PET, although spectrally different, are not unlike those observed here for 4-CBA. This will form the next stage in our programme of work as well as phosphorescence analysis.

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